



UNIVERSITY OF COLOMBO, SRI LANKA

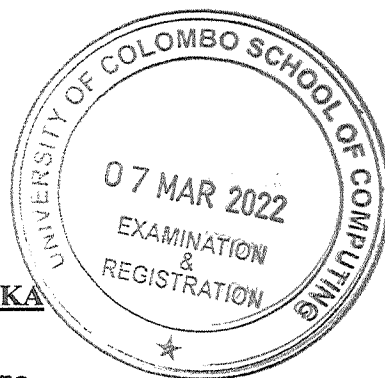
University of Colombo School of Computing

BACHELOR OF SCIENCE IN INFORMATION SYSTEMS

Second Year Examination – Semester I – 2021/2022

IS2104 – Rapid Application Development (Part B)

TWO (2) HOURS (for both parts A & B)



97

To be completed by the candidate

Examination Index No:

Important Instructions to candidates:

1. The medium of instruction and question is **English**.
2. Write your answers in **English**.
3. If a page or a part of this question paper is not printed, please inform the supervisor immediately.
4. Note that questions appear on both sides of the paper. If a page is not printed, please inform the supervisor immediately.
5. Write your index number on each and every page of the answer paper.
6. Part B has 2 questions in **08** pages.
7. Answer **ALL** questions. All questions carry equal marks (**25** marks).
8. **This paper consists of two parts, Part A (Question No 1 and Question No 2) and Part B (Question No 3 and Question No 4) and submit separately.**
9. Any electronic device capable of storing and retrieving text including electronic dictionaries and mobile phones are not allowed.
10. **Non-Programmable/Programmable** calculators are **not allowed**.

For Examiner's use only

Question No	Marks
3	
4	
Total	

Index No:

Part B

Question 3

- (a) Briefly describe the difference between “Interface” and “Abstract Class”?

[2 Marks]

- (b) Explain what is “Encapsulation in Java”.

[3 Marks]

Index No:

- (c) Explain method overloading and method overriding using a suitable example in Java language. [4 Marks]

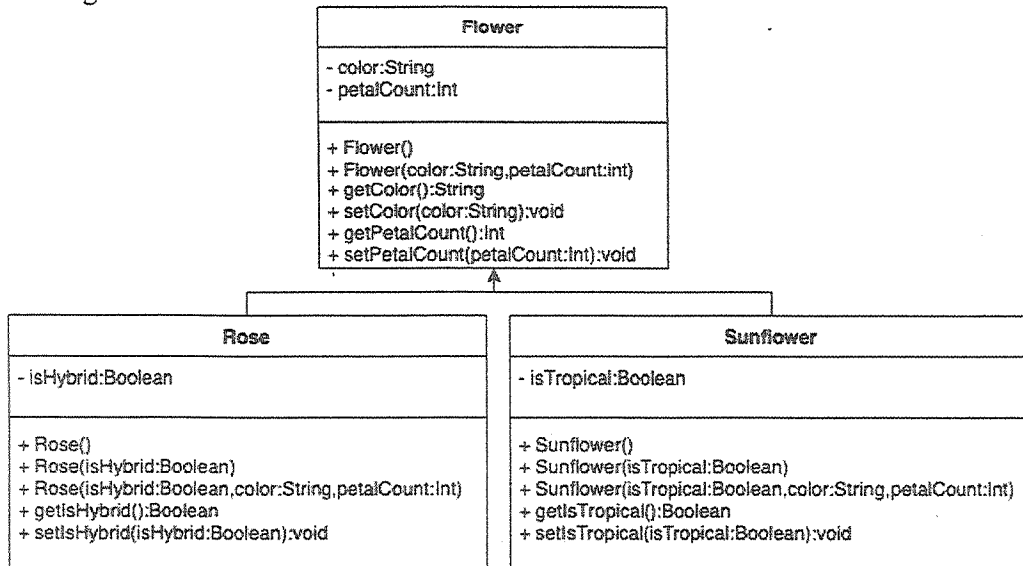
- (d) Explain the following access modifiers.

- Default
- Private
- Protected

[6 Marks]

Index No:

- (e) Write down the Java Code for the “Flower” superclass and “Rose” subclass, shown in the below class diagram.



[10 Marks]

4

Index No:

- (b) List two (2) examples for runtime errors.

[4 Marks]

- (c) A java code segment and its output are shown below. Complete the lines 5,8,14,15,22, and 23, and write the completed code lines in the table provided below.

[6 marks]

Code:

```
1 public class RAD {
2     public void printPattern() {
3         String printChar;
4
5         for (_____ ; i < 11; i++) {
6             String printLine="";
7
8             if (_____ % _____ == _____) {
9                 printChar = "*";
10            } else {
11                printChar = "#";
12            }
13
14            for(_____ ;j<=i;j++){
15                _____ = _____.concat(_____);
16            }
17            System.out.println(printLine);
18        }
19    }
20
21    public static void main(String args[]) {
22        _____;
23        _____;
24    }
25 }
```

Output :

```

*
##
***
####
*****
#####
*****
#####
*****
#####

```

Line No	Code Line
5	
8	
14	
15	
22	
23	

- (d) List three (3) compile-time errors with one (1) example each and mention the error in each example as a comment in the code

[6 Marks]

```

public class rad{
    public static void main(String args[]) {
        // Your example code here
    }
}

```


Index No:

- (e) List three (3) ways to make an object eligible for java garbage collection. Give one (1) example for each method.

[6 Marks]
